2025 IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS) | 979-8-3315-9172-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/BCICTS63111.2025.11211162

Verification of compact model nonlinearity using an H-band frequency quadrupler fabricated in SG13G3Cu Technology

Boli Peng[1], Markus Müller[2], Michael Schröter[1]
[1]Chair for Electron Devices and Integrated Circuits (CEDIC), TU Dresden, 01062 Germany
[2]formerly with CEDIC, now with Rohde&Schwarz, Munich, Germany
Corresponding author: boli.peng@tu-dresden.de

***Abstract* - A detailed device-centered analysis is performed for a frequency quadrupler that has been fabricated in the presently most advanced SiGe heterojunction bipolar transistor (HBT) process technology. The analysis serves for verifying the non-linear behavior of an advanced compact model and provides to model developers important feedback on the impact of device parameters and physical effects on circuit performance.**

Keywords: compact modeling, frequency multiplier, heterojunction bipolar transistors, non-linear modeling

I. Introduction

For enabling high-frequency (HF) 90-300 GHz applications, such as wireless communications [1], ViSAR for airborne radar [2], and hyperspectral imaging [3], sub-terahertz (sub-THz) signal generation with large bandwidth (BW) and sufficient output power for signal transmission is required. Compared to voltage controlled oscillators, frequency multipliers are preferred for sub-THz signal generation, as they not only provide broadband signals with lower phase noise but also higher output power through the conversion gain of the multiplier.

Reliable predictions of the performance of fabricated HF circuits during the design phase requires accurate active and passive device models in the employed process design kit (PDK). Model accuracy is determined by (i) the model formulation itself, i.e. the coverage of all relevant physical effects occurring in a device, and (ii) the model parameter values that are extracted from experimental data. Today's industry PDKs include sophisticated compact models (CMs), especially for transistors which are the most complicated structures in integrated circuits. With progressing device size reduction, more and more physical effects, especially regarding carrier transport, come into play that make CM development and verification challenging. Of particular interest for almost all circuits is the accurate description of the non-linear transistor behavior.

Non-linear CM behavior is often investigated and tested based on load-pull measurements. However, the data are limited to lower frequencies (typically 67GHz) and may not cover the operating conditions that a transistor encounters in a circuit. Hence, verifying CMs via experimental circuit performance data provides direct and additional confirmation of its suitability in practical applications and important confirmation on relevant effects. Unfortunately, the circuit design community provides very little feedback regarding the impact of device parameters and physical effects on circuit performance to foundries and model developers. Therefore, this work presents a detailed device-centered analysis of an H-band frequency quadrupler [4] that has been fabricated in the presently most advanced SiGe HBT process technology and relies on the non-linear behavior of the HBTs. The circuit is shown in Fig. 1.

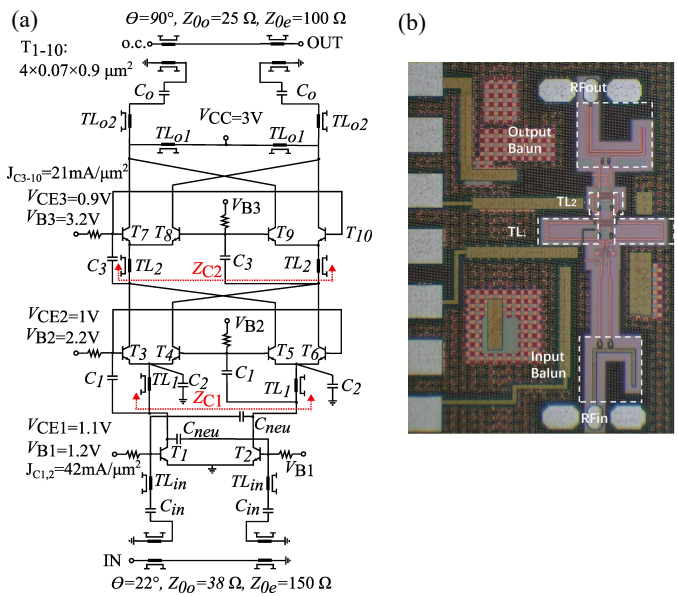


Fig. 1: (a) Schematic and (b) chip micrograph of the fabricated stacked Gilbert cell frequency quadrupler. For the HBTs, four parallel BEC unit-cells are used in all stages.

II. Process Technology and Device Modeling

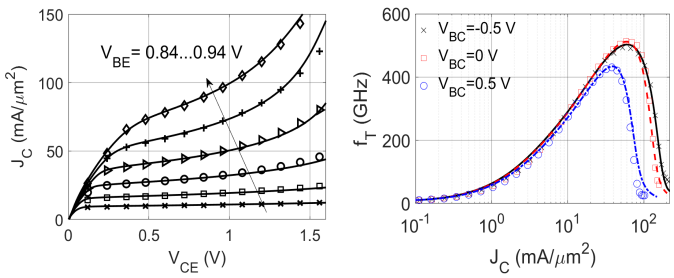
The frequency quadrupler was implemented in IHP's highest performance 130-nm SiGe BiCMOS technology SG13G3Cu, featuring HBTs with (f_T, f_{max}) values of (470, 650) GHz [5, 6]. The PDK offers HICUM/L2 v2.4.0 [7] parameters for performance optimized HBT unit-cells with BEC configuration and an emitter size of 0.07 µm×0.9 µm. Measured output characteristics and transit frequency are shown in Fig. 2. The compact model agrees quite well with the experimental data in the relevant bias range after adjusting few parameters to the data measured on the same dies that contain the circuits.

III. Multiplier Circuit

Fig. 1 shows the circuit schematic and corresponding chip micrograph of the H-band frequency quadrupler. Fed with an input signal of frequency f_{in} = 55-80 GHz, its core consists of two stacked bootstrapped Gilbert-cell (GC) doublers to generate an output voltage signal with a time period of $1/(4f_{in})$. The corresponding 232-282 GHz output signal is obtained after a bandpass filter. A detailed discussion of the circuit can be found in [4].

979-8-3315-9172-4/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on April 01,2026 at 12:12:53 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 2: (a) Output characteristics and (b) transit frequency of a 4×0.07×0.9 μm² BEC HBT. Comparison between model (lines) and measurement (symbols).

A. *Circuit figures of merit*

Analysis of the circuit in Fig. 1(a) and assuming that the input impedance seen into the emitter of each quad stage equals Z_{C1} and Z_{C2} gives for the output power

$$P_{out} = \frac{R_{L,opt}}{2}\left(\frac{g_{m,eff}^4 Z_{C1}^2 Z_{C2} V_{in}^4}{64 m_C^3 V_T^3}\right)^2 \quad (1)$$

and for the (power) conversion gain

$$CG = R_{L,opt} R_{S,opt}\left(\frac{g_{m,eff}^4 Z_{C1}^2 Z_{C2} V_{in}^3}{64 m_C^3 V_T^3}\right)^2 . \quad (2)$$

Here, $R_{L,opt}$ and $R_{S,opt}$ are the optimum load and source resistance determined through large-signal source- and load-pull *simulations*, V_{in} is the differential voltage amplitude at the g_m-stage input, m_C is the collector current non-ideality factor in the operating point (assuming $I_C = I_S \exp(V_{BE}/(m_C V_T))$), and

$$g_{m,eff} = \frac{g_{mi,0}}{\sqrt{D + (\omega(R_B + R_E)[C_{BE} + C_{BC}(1 + g_{mi,0} R_L)])^2}} \quad (3)$$

with $D = (1+g_{mi,0}R_E)^2$ is the device effective transconductance, including the effect of the input time constant ($R_B C_{BE}$), emitter resistance (R_E) and Miller effect (C_{BC}); $g_{mi,0}$ is the internal g_m in the operating point.

Above equations have been derived under the assumption of a sufficiently small input signal amplitude ($\leq V_T$) and hence loose their validity at high input power. Nevertheless, they are still useful for understanding the circuit operation and for identifying the relevant transistor parameters. Both g_m and $Z_{C1,2}$ enter the P_{out} and CG with high order, indicating that their accurate modeling is crucial for improving the agreement between measurement and simulation.

B. *Circuit Measurement and Comparison to Simulation*

A PNA-X was used to generate the low frequency input signal ranging from 55-67 GHz, while the 67-80 GHz signal was generated through an Agilent N5260-60003 67-110 GHz extender. The output 4th harmonic power was measured with an H-band harmonic mixer RPG HM 220-330 and R&S FSW67 spectrum analyzer. By subtracting the insertion losses of the input I110-GSG-100 probe, output 325B-GSG-80-BT probe and connecting waveguides and cables according to their datasheet specifications, their effect on the circuit measurements has been de-embedded. These losses are very close to those measured directly using a VNA. However, the impact of reflections due to possible frequency dependent deviations of the probe impedance from 50 Ω as well as of possible radiation losses at high frequencies and crosstalk between adjacent probes on measured data have not been removed.

Fig. 3 shows the measured output power versus input power at (236, 252, 276) GHz. Given the remaining measurement errors (including about +/-5% in the power measurement) and fabrication tolerances, the simulation using HICUM/L2 with nominal model parameters agrees quite well. However, not including the parasitic capacitances (C_{par}) of the HBT cell leads to ≈25 dBm increase in P_{out} at low P_{in}. At high P_{in} self-heating (SH) reduces P_{out} up to 3dB, but its impact is limited due to the external base biasing resistance and relatively small even harmonics level. High-current effects (HC) have no visible impact.

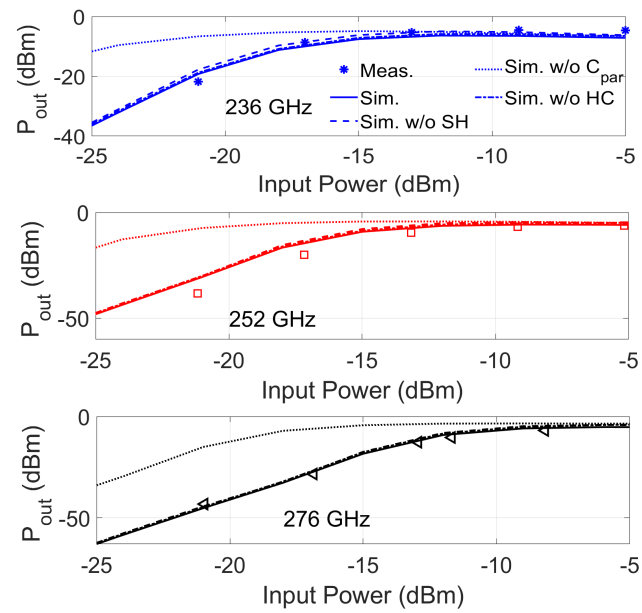


Fig. 3: Measured (symbols) and simulated (lines) output power versus input power at different output frequencies. Note the compressed P_{out} scale. Labels: SH (self-heating), HC (high-current effects), C_{par} (parasitic capacitances).

Fig. 4 shows the measured conversion gain at input power levels of -20 dBm and -8 dBm. Using the nominal model parameters, the simulated CG agrees reasonably well with the measurement. For both power levels the deviations increase towards higher frequencies. The influence of C_{par} and SH is clearly visible.

C. *Impact of key passive devices*

Circuit performance is influenced by both transistor parameters and passive elements. To obtain a feel for the relevance of the latter it is instructive to evaluate their impact on circuit performance. The key passives here, according to the analysis,

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on April 01,2026 at 12:12:53 UTC from IEEE Xplore. Restrictions apply.

are the transmission lines TL1 and TL2. During design the electrical length of TL1 (and TL2) is optimized for the center frequency, but changes with frequency during operation. This leads to a change of the impedances Z_{C1} and Z_{C2} seen into the emitter nodes of the quad stages, which in turn changes the dependence of CG and P_{out} according to (1) and (2).

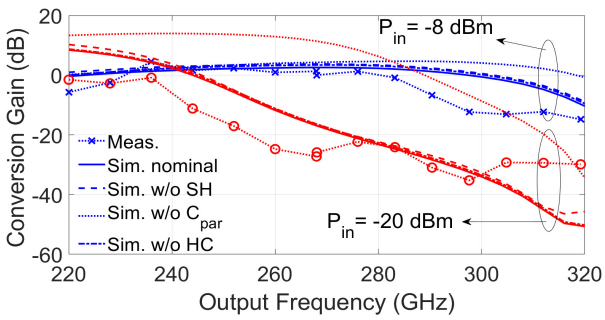


Fig. 4: Measured (symbols) and simulated (lines) conversion gain of the quadrupler at P_{in} = −8 dBm and -20 dBm.

Fig. 5 (a) shows that a 20% larger (smaller) electrical length of TL1 causes a decrease (increase) of up to 7 dB in CG at high P_{in} and >3dB over the entire frequency range. In contrast, for low input power, CG changes by maximal 3dB at just the upper frequency limit. The impact of TL2 is much smaller and not shown here.

The other important passive device is the neutralization capacitance C_{neu} in the g_m-stage, which also leads to a change in Z_{C1} and Z_{C2}. With its small value of 12 fF, it is expected to be quite vulnerable to process variations. Fig. 5 (b) shows that a 20% variation has a very similar impact on CG as TL1.

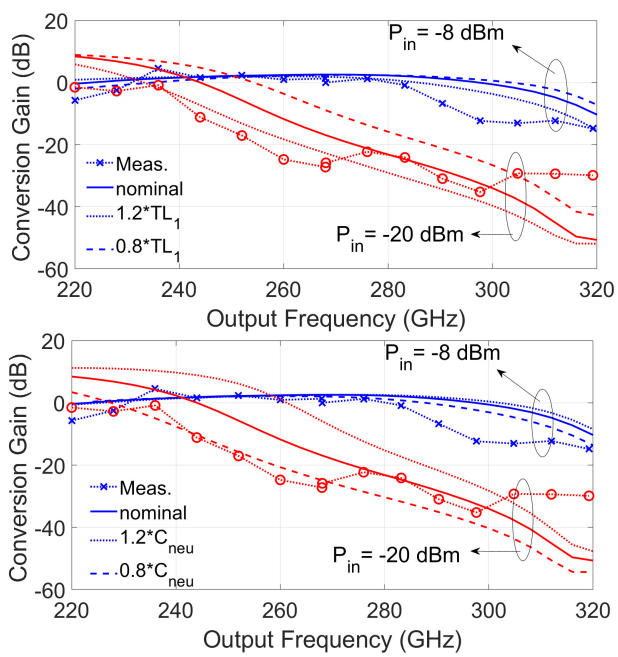


Fig. 5: Measured (symbols) and simulated (lines) conversion gain at P_{in} = (−8, -20) dBm, displaying the impact of 20% variation of (a) TL1 length and (b) neutralization capacitance value.

D. *Transistor Parameter Sensitivity Study*

The purpose of this sensitivity study is to show the impact of variations in device parameters on circuit performance and to identify the most relevant model parameters and physical effects for the investigated circuit. The study was performed by varying selected basic model parameters of HICUM/L2 within the ranges estimated for process variations in device characteristics. The correlation between some of the parameters was neglected due to, among others, the unavailability of information on process parameter variations.

Fig. 6(a) displays the impact of those model parameters that turned out to be most relevant for CG at low input power. The largest shift in the drop of CG (by over 40 GHz) results from a lower BC depletion capacitance (C_{jCi0}), followed by lowering the mobile carrier transit time τ_0 (t0) with a shift of ≈20 GHz. Somewhat unexpectedly, a lower internal base current (ibeis) causes a visible drop of CG. This can be explained by the lower voltage drop across the biasing resistors of the quad-stages, which leads to a slightly higher V_{BE} and thus a higher I_C and g_m. The latter decreases the impedance seen into emitter of the quad stage transistors, changing the inter-stage matching. This leads to a lower voltage and overall conversion gain, since the current swings remain about the same. Note that a higher g_m also increases BE diffusion capacitance and hence the charging current component. Hence, accurately modeling g_m and the elements entering $g_{m,eff}$ in (3) is key for predicting the bandwidth correctly.

At high input power the results, shown in Fig. 6(b), look quite different. A shift of the collector operating current $I_{C,op}$ and thus g_m (via either c10 or ibeis) has the largest effect on CG over the entire BW. At the high frequency end, both τ_0 and C_{jCi0} have a similar impact. In contrast, the thermal resistance (rth) and emitter resistance (re) as well as the external base and collector resistance (rbx, rcx) have a relatively small influence on CG at high input power.

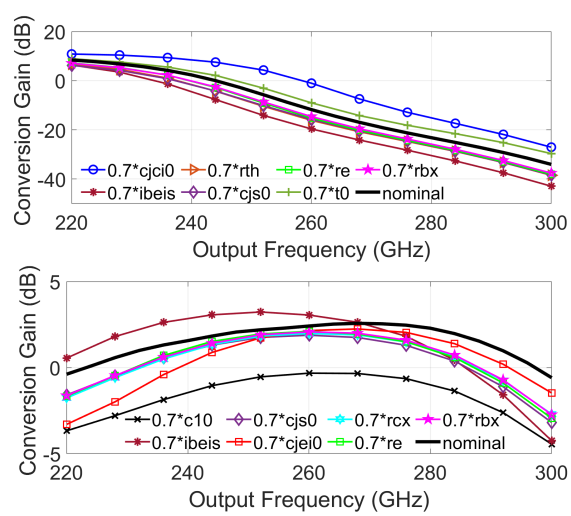


Fig. 6: Change in CG over 220-300 GHz obtained from simulations at (a) P_{in} = -20 dBm and (b) P_{in} = -8 dBm with device parameter reduction of 30%. Note the large difference in *y* axis scales.

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on April 01,2026 at 12:12:53 UTC from IEEE Xplore. Restrictions apply.

Fig. 7 displays the output power as a function of transistor parameter changes for the lower and upper limit of the BW as well as for the center frequency. In all cases, a change in quiescent collector current (via c10 and ibeis) has the largest impact, followed by the depletion capacitances (cjci0, cjei0) and transit time (t0). The relative impact of the charge storage elements differs depending on the operating frequency. High-frequency related effects such as a distribution of the BC and BE capacitance across the base resistance and non-quasi-static effects [7] only play a minor role.

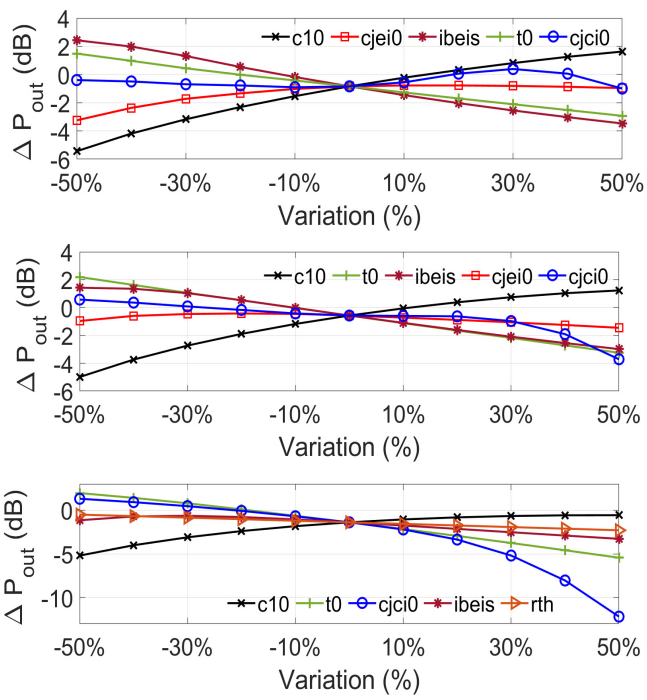


Fig. 7: Change in output power at (a) 236 GHz, (b) 252 GHz and (c) 276 GHz obtained from simulations at P_{in} = -8 dBm with device parameter variation up to 50%.

E. *Time domain analysis*

Fig. 8 shows selected results of the time domain behavior of the transistors, which reveals the most detailed information on the operating ranges in the circuit that need to be accurately modelled. For P_{in} = -8 dBm, the peak of the HBT (quasi-static) transfer current density $j_T(t)$ reaches almost 100 mA/µm^2 at an internal CE voltage $v_{CEi}(t) \approx 0.6$ V in the input stage. At this slightly positive BC voltage $f_{T,peak}$ is exceeded only slightly. In the upper stage, the peak $j_T(t)$ of 60 mA/µm^2 with $v_{CEi}(t) \approx 0.7$ V never exceeds peak f_T. A minimum value of 0.4 V for $v_{CEi}(t)$ is reached in the upper quad stage but at a very low current density. Thus, overall, high-current effects do not play a significant role, confirming the sensitivity analysis results above.

On the high voltage side, peak $v_{CEi}(t)$ of 1.1 V remains below the static breakdown voltage BV_{CEO} (≈ 1.5 V) in the upper quad stage, while the peak $v_{CEi}(t) = 1.9$ V in the input stage significantly exceeds BV_{CEO} but at almost zero j_T. This does not lead to degradation though due to the short time the transistors spend in such states during dynamic operation [8].

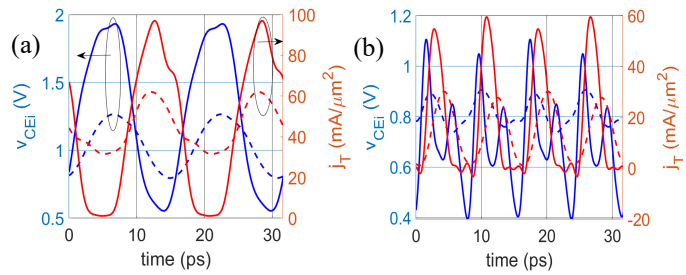


Fig. 8: Time dependent transfer current density (red lines) and internal CE voltage (blue lines) at 252 GHz for transistors in (a) the input stage and (b) the upper quad stage for P_{in} = -20 dBm (dashed lines) and P_{in} = -8 dBm (solid lines).

IV. Summary and Conclusions

The suitability of an advanced compact model for practical applications has been demonstrated based on a frequency quadrupler fabricated in the presently most advanced SiGe HBT technology. A detailed device-centered analysis identifies the most relevant circuit elements and physical effects, including unexpected ones (according to theory), impacting circuit performance. The results provide valuable feedback not only for model developers and foundries but also to circuit designers regarding the potential causes for the often observed deviations between circuit simulation and measurement.

Acknowledgment

Partial funding from the German National Science Foundation (DFG grants SCHR695/14, /27,/ 29) is acknowledged.

References

[1] M. Rodwell et al., “100-300GHz Wireless: ICs, Arrays, and Systems”, IEEE BCICTS, Monterey, CA, USA, pp. 1-4, 2021.

[2] S. -H. Kim, R. Fan and F. Dominski, “ViSAR: A 235 GHz radar for airborne applications”, IEEE Radar Conf. (RadarConf), Oklahoma City, OK, USA, pp. 1549-1554, 2018.

[3] T. Chi, M. -Y. Huang, S. Li and H. Wang, “A packaged 90-to-300GHz transmitter and 115-to-325GHz coherent receiver in CMOS for full-band continuous-wave mm-wave hyperspectral imaging”, IEEE ISSCC, San Francisco, CA, USA, pp. 304-305, 2017.

[4] B. Peng and M. Schröter, “A 232-282 GHz Frequency Quadrupler in 130-nm SiGe HBT SG13G3Cu Technology”, IEEE Microw. Comp. Lett., acc. for public., 2025.

[5] B. Heinemann et al., "SiGe HBT with fT/fmax of 505 GHz/ 720GHz", IEDM Tech. Digest, pp. 51-54, 2016.

[6] H. Rücker and B. Heinemann, “High-performance SiGe HBTs for next generation BiCMOS technology,” Semicond. Sci. Technol., vol. 33, no. 11, pp. 114003, 2018.

[7] M. Schröter and A. Chakravorty, “Compact hierarchical modeling of bipolar transistors with HICUM”, World Scientific, Singapure, 2010.

[8] C. Weimer, G. Fischer and M. Schröter, “Characterization, Analysis and Modeling of Long-Term RF Reliability and Degradation of SiGe HBTs for High Power Density Applications”, IEEE Trans. Dev. Mat. Reliab., vol. 24, no. 1, pp. 20-32, 2024.

Authorized licensed use limited to: FUDAN UNIVERSITY. Downloaded on April 01,2026 at 12:12:53 UTC from IEEE Xplore. Restrictions apply.